Note on a special Matrix

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Abstract

Analysis of special matrix

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1. Some analysis of a special symmetric matrix

In this note I will be examining a special type of symmetric matrix with constant diagonal and all the off diagonal entries the same but different from the main diagonal. Below is the matrix spoken of:

$$\mathbf{M}_n = \begin{bmatrix} a & b & b & b & \cdots & b & b \\ b & a & b & b & \cdots & b & b \\ b & b & a & b & \cdots & b & b \\ b & b & b & a & \cdots & b & b \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ b & b & b & b & \cdots & a & b \\ b & b & b & b & \cdots & b & a \end{bmatrix}, \tag{1}$$

where $a, b \in \mathbb{H}$; thus, $\mathbf{M}_n$ is a square $n \times n$ symmetric matrix. My goal is to find the determinant of such a matrix, I claim that $\det(\mathbf{M}_n) = (a-b)^{n-1}(a+(n-1)b)$. I will also define the following special matrices, the use of which will become apparent later:

$$\mathbf{M}'_n = \begin{bmatrix} a & b & b & b & \cdots & b & b \\ a-b & b-a & 0 & 0 & \cdots & 0 & 0 \\ a-b & 0 & b-a & 0 & \cdots & 0 & 0 \\ a-b & 0 & 0 & b-a & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ a-b & 0 & 0 & 0 & \cdots & b-a & 0 \\ a-b & 0 & 0 & 0 & \cdots & 0 & b-a \end{bmatrix}. \tag{2}$$

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This matrix $\mathbf{M}'_n$ is obtained from $\mathbf{M}_n$ by subtracting each row of $\mathbf{M}_n$ from the first row of $\mathbf{M}_n$. Another way to think of this is multiplying every row after the first by -1, then adding it to the first row, and then that replaces the current row. In other words

$$\mathbf{M}_n \xrightarrow{r'_j = r_1 + (-1)\cdot r_j} \mathbf{M}'_n,$$

for every $j \in [2, \ldots, n]$. As a result of the rows operations the determinants of $\mathbf{M}_n$ and $\mathbf{M}'_n$ are related in the following way:

$$\det(\mathbf{M}_n) = (-1)^{n-1}\det(\mathbf{M}'_n). \tag{3}$$

This is because for every row multiplied by -1 the determinant is multiplied by -1; therefore, $n-1$ rows are being multiplied by -1 and this causes the determinant to be multiplied by $(-1)^{n-1}$. Adding the rows does not affect the determinant.

Next, I will define the special matrix below

$$\mathbf{S}_n = \begin{bmatrix} b & b & b & \cdots & b & b \\ b-a & 0 & 0 & \cdots & 0 & 0 \\ 0 & b-a & 0 & \cdots & 0 & 0 \\ 0 & 0 & b-a & \cdots & 0 & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & \cdots & b-a & 0 \end{bmatrix}. \tag{4}$$

Notice that we can easily calculate the determinant of $\mathbf{S}_n$ by using the Laplacian determinant expansion along the bottom row. As a side note, $n + (n-1)$ is the sum of two consecutive integers, this is always an odd number. Therefore, $(-1)^{(n-k)+(n-(k+1))} = -1$. If we couple this with recursion we get $\det(\mathbf{S}_n) =$

$$= (b-a)(-1)^{n+(n-1)}\det(\mathbf{S}_{n-1}) = (b-a)(-1)^1\det(\mathbf{S}_{n-1})$$

$$= (b-a)^2(-1)^{n+(n-1)}(-1)^{(n-1)+(n-2)}\det(\mathbf{S}_{n-2}) = (b-a)^2(-1)^2\det(\mathbf{S}_{n-2})$$

$$= (b-a)^3(-1)^{n+(n-1)}(-1)^{(n-1)+(n-2)}(-1)^{(n-2)+(n-3)}\det(\mathbf{S}_{n-3}) = (b-a)^3(-1)^3\det(\mathbf{S}_{n-3})$$

$$= (b-a)^q(-1)^q\det(\mathbf{S}_{n-q}),$$

where q is the recursive depth of the calculation. Since $\mathbf{S}_n$ only has $n-1$ entries of $b-a$, this means that we can only calculate up to a recursive depth of $q = n-1$ and we will *reach* $\mathbf{S}_1 = [b]$. If we take the recursion all the way to $q = n-1$ we get

$$\begin{aligned} \det(\mathbf{S}_n) &= (b-a)^{n-1}(-1)^n\det(\mathbf{S}_1) \\ &= (b-a)^{n-1}(-1)^n \cdot b. \end{aligned}$$

Hence it follows

$$\det(\mathbf{S}_n) = b(a-b)^{n-1}. \tag{5}$$

1.1. Determinant of the special matrix for 2 by 2 case

In this small section I will examine the 2×2 of the matrix described by (1). Let the 2×2 version of (1) be

$$\mathbf{M}_2 = \begin{bmatrix} a & b \\ b & a \end{bmatrix}.$$

The corresponding form of the row reduced matrix (2) is given by

$$\mathbf{M}_2' = \begin{bmatrix} a & b \\ a-b & b-a \end{bmatrix}.$$

Notice that since we obtain $\mathbf{M}_2'$ by multiplying the second row of $\mathbf{M}_2$ by -1 and then adding row 1 and row 2 back into row 2, this affects the determinants in the following way:

$$\det(\mathbf{M}_2) = (-1)^1 \det(\mathbf{M}_2').$$

This follows (3) and is easily verified:

$$\det(\mathbf{M}_2) = a^2 - b^2 = (a-b)(a+b),$$

and

$$\det(\mathbf{M}_2') = a(b-a) - b(a-b) = -(a-b)(a+b); \tag{6}$$

therefore, the determinants differ by one multiple of -1. Also notice that $\det(\mathbf{M}_2)$ is of the form claimed at the beginning:

$$(a-b)(a+b) = (a-b)^{2-1}(a+(2-1)b)$$

1.2. Determinant of the special matrix for 3 by 3 case

In this section I will calculate the determinant of the special matrix $\mathbf{M}_n$, following similar analysis for the 2×2 case. Let the 3×3 version of the matrix (1) and its row reduced version (2) be as follows:

$$\mathbf{M}_3 = \begin{bmatrix} a & b & b \\ b & a & b \\ b & b & a \end{bmatrix} \rightarrow \mathbf{M}_3' = \begin{bmatrix} a & b & b \\ a-b & b-a & 0 \\ a-b & 0 & b-a \end{bmatrix}.$$

The same rules of determinants and row operations affect how the determinants are related and we exploit (3) to get

$$\det(\mathbf{M}_3) = (-1)^2 \det(\mathbf{M}_3'). \tag{7}$$

Once we know $\det(\mathbf{M}_3')$ then we can find $\det(\mathbf{M}_3)$. We can use the Laplacian determinant expansion along the bottom row of $\mathbf{M}_3'$ to get

$$\det(\mathbf{M}_3') = (a-b)(-1)^{3+1} \begin{vmatrix} b & b \\ b-a & 0 \end{vmatrix} + (b-a)(-1)^{3+3} \begin{vmatrix} a & b \\ a-b & b-a \end{vmatrix}. \tag{8}$$

Notice that the two matrices on the right hand side of (8) are of the forms (4) and (2) respectively. If we utilize (5) and (6) then (8) becomes

$$\begin{aligned}\det(\mathbf{M}'_3) &= (a-b)\det(\mathbf{S}_2) + (b-a)\det(\mathbf{M}'_2) \\ &= (a-b)\cdot b(a-b)^{2-1} + (b-a)\cdot(-(a-b)(a+b)) \\ &= (a-b)^2(b+(a+b)) \\ &= (a-b)^2(a+2b).\end{aligned}$$

By (7) we have

$$\det(\mathbf{M}_3) = (a-b)^2(a+2b).$$

Notice that this verifies our claim at the beginning for the form of the determinant

$$\det(\mathbf{M}_3) = (a-b)^{3-1}(a+(3-1)b).$$

1.3. Determinant of the special matrix for the general case

If we calculate the Laplacian determinant expansion along the bottom row of the matrix (2) we get

$$\det(\mathbf{M}'_n) = (a-b)(-1)^{n+1}\det(\mathbf{S}_{n-1}) + (b-a)\det(\mathbf{M}'_{n-1}). \tag{9}$$

As another side trick, notice that $(-1)^{n+1} = (-1)^{n-1}$, then if we substitute (5) (where in (5) n goes to $n-1$) into (9) we get

$$\begin{aligned}\det(\mathbf{M}'_n) &= (a-b)(-1)^{n+1}\cdot b(a-b)^{n-2} + (b-a)\det(\mathbf{M}'_{n-1}) \\ &= b(a-b)^{n-1}(-1)^{n+1} + (b-a)\det(\mathbf{M}'_{n-1}) \\ &= b(a-b)^{n-1}(-1)^{n-1} + (b-a)\det(\mathbf{M}'_{n-1}) \\ \det(\mathbf{M}'_n) &= b(b-a)^{n-1} + (b-a)\det(\mathbf{M}'_{n-1}) \qquad (10)\end{aligned}$$

We can calculate $\det(\mathbf{M}'_{n-1})$ and get

$$\det(\mathbf{M}'_{n-1}) = b(b-a)^{n-2} + (b-a)\det(\mathbf{M}'_{n-2}). \tag{11}$$

If we substitute (11) back into (10) we get

$$\begin{aligned}\det(\mathbf{M}'_n) &= b(b-a)^{n-1} + (b-a)\cdot(b(b-a)^{n-2} + (b-a)\det(\mathbf{M}'_{n-2})) \\ &= 2b(b-a)^{n-1} + (b-a)^2\det(\mathbf{M}'_{n-2}),\end{aligned}$$

and this would be the calculation to a recursive depth of 2. If we repeat the recursion again, depth of 3, we get

$$\det(\mathbf{M}'_n) = 3b(b-a)^{n-1} + (b-a)^3\det(\mathbf{M}'_{n-3}).$$

In general at a recursive depth of q we get

$$\det(\mathbf{M}'_n) = qb(b-a)^{n-1} + (b-a)^q \det(\mathbf{M}'_{n-q}). \tag{12}$$

Notice that $\mathbf{M}'_n$ has only $n-1$ diagonal entries of $b-a$; thus we can only take the recursion to a depth $q = n-1$ before we *reach* the matrix $\mathbf{M}'_1 = [a]$. If we set $q = n-1$ in (12) we get

$$\begin{aligned}
\det(\mathbf{M}'_n) &= (n-1) \cdot b(b-a)^{n-1} + (b-a)^{n-1} \det(\mathbf{M}'_1) \\
&= (n-1) \cdot b(b-a)^{n-1} + a(b-a)^{n-1} \\
\det(\mathbf{M}'_n) &= (b-a)^{n-1}(a + (n-1)b). \qquad (13)
\end{aligned}$$

If we substitute (13) into (3) we get

$$\begin{aligned}
\det(\mathbf{M}_n) &= (-1)^{n-1} \cdot (b-a)^{n-1}(a + (n-1)b) \\
&= (a-b)^{n-1}(a + (n-1)b), \qquad (14)
\end{aligned}$$

this verifies our claim from before.